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AI-driven dynamic trust management and blockchain-based security in industrial IoT

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ABSTRACT

The Industrial Internet of Things (IIoT) revolutionizes industrial operations through real-time data exchange and analytics but introduces significant security and trust challenges particularly in dynamic and distributed IIoT environments. We hypothesize that integrating an AI-driven Trust Management System (TMS) with blockchain technology can address these issues effectively. This paper proposes a framework combining an AI-driven Dynamic TMS (AI-DTMS) with a private blockchain. AI-DTMS evaluates the reliability of the device and data using machine learning, achieving 96.31% accuracy with minimal false positives. The blockchain module ensures secure authentication, achieving nearly 100% success. It mitigates critical threats, including spoofing, Sybil, node-capturing, replay, and DDoS attacks, ensuring robust security in IIoT environments. Performance evaluations demonstrate 35% improvement in response time and up to 97.8% reduction in latency, underscoring scalability and efficiency. By integrating AI-DTMS with blockchain, the framework enhances trust, security, and performance in dynamic IIoT environments, offering a scalable and robust solution.

1. Introduction

In recent years, the Industrial Internet of Things (IIoT) has transformed industrial operations, enhancing connectivity and enabling real-time data exchange [1]. IIoT systems foster automation, monitoring, and predictive maintenance, driving efficiency and innovation. However, the widespread deployment of connected devices introduces significant challenges in data security, reliability, and resource-efficient communication [2]. Many IIoT devices lack essential security features like encryption and authentication, while legacy systems fail to meet modern standards, increasing vulnerability to attacks [3]. Moreover, manual lifecycle management of IIoT devices is resource-intensive, and the vast amount of data exchanged raises privacy concerns.

Let us consider an automotive factory deploying IIoT devices like sensors, actuators, and cameras to optimize production. These devices generate continuous data on machine performance, environmental conditions, and product specifications. Trust in the accuracy and security of this data is crucial to prevent production errors, ensure safety compliance, and protect intellectual property. As IIoT integration grows, malfunctions, security breaches, and scalability issues become more prevalent, emphasizing the importance of strong trust and security systems.

Addressing these vulnerabilities requires not only security measures but also systems to assess the reliability of devices and interactions within the network. A Trust Management System (TMS) can mitigate risks by dynamically evaluating device behavior, reducing uncertainty in interactions with potentially malicious entities [4]. Trust and security are especially critical in IIoT environments, where devices interact dynamically in harsh, unregulated settings. Ensuring secure data exchange and trusted devices

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is vital to maintaining the integrity of industrial systems [5]. Traditional security measures are often insufficient for the dynamic and distributed nature of IIoT networks, necessitating the development of more robust and adaptive solutions [6].

Further, blockchain technology, initially recognized for cryptocurrency applications like Bitcoin, offers immutability and transparency. Researchers expanded its applications to domains such as IIoT and Healthcare IoT, introducing features like Smart Contracts, access control [7], modularity, scalability, etc. For example, permissioned blockchain such as Hyperledger Fabrics (HF) is designed and developed for industrial applications where more control transactions, smart decisions, and privacy are more important. Leveraging blockchain and Artificial Intelligence (AI), recent architectures have demonstrated improved security and scalability in smart manufacturing environments [8,9].

The key contributions of the paper include the development of a novel approach combining an AI-driven Dynamic TMS (AI-DTMS) with a private blockchain-based security framework. The AI-DTMS employs machine learning to dynamically evaluate device trustworthiness, ensuring secure interactions. Concurrently, the blockchain framework, leveraging the Elliptic Curve Digital Signature Algorithm (ECDSA) and smart contracts, strengthens device authentication and data integrity. The proposed system demonstrates enhanced scalability, security, and reliability compared to traditional methods, making it a robust solution for securing IIoT environments. The remaining paper is organized as follows: Section 2 reviews related work, identifying gaps addressed by the proposed system. Section 3 explains the methodology and architecture of AI-DTMS. Section 4 describes the integration of HF blockchain for IIoT security. Section 5 outlines system implementation, and Section 6 analyzes the results of the proposed system.

2. Literature review

Several studies have explored TMS and security mechanisms in IIoT environments, each addressing specific aspects while leaving certain gaps. For instance, [10] proposes an elliptic curve-based authentication mechanism with low computational cost but lacks scalability validation for larger networks. Similarly, [11] explores IoT authentication using Ethereum blockchain but requires scalability assessment. Study [12] demonstrates a decentralized authentication mechanism for lightweight IoT devices with limited broader validation, while [13] uses blockchain-enabled fog nodes for user authentication, offering scalability benefits needing further evaluation.

An AI-enabled TMS (AIT) [14] combines blockchain and deep learning for vehicular networks but suffers from centralized design limitations in dynamic IIoT environments. A decentralized protocol [15] leverages blockchain for transparency but relies heavily on historical data, inadequate for evolving IIoT systems. Incorporating advanced machine learning could address this. Another architecture [9] integrates blockchain for data integrity and machine learning for attack detection, effectively countering DDoS and XSS attacks but lacking dynamic trust management. Similarly, a voting-based access control system [16] ensures robust authorization but misses scalable trust mechanisms in heterogeneous IIoT settings. The framework [5] employs a Proof of Authority consensus mechanism for energy-efficient security but lacks scalability and dynamic trust evaluation for real-time IIoT. A blockchain-based architecture [17] introduces a cloud–edge-end cooperative network addressing data security but lacks adaptability to dynamic environments. The main-secondary chain framework [18] improves throughput in multi-domain IIoT using trust tickets but falls short on real-time adaptability. A federated learning-based framework [19] tackles cross-domain trust with privacy innovations but lacks scalable architecture for real-time management.

In summary, while these studies advance IIoT trust management, they leave gaps in scalability, adaptability, and integration of advanced methodologies. This research addresses these gaps through the following questions (RQ): RQ1: How can a TMS with blockchain enhance accuracy, security, and adaptability in IIoT while minimizing biases and trade-offs? RQ2: How can the framework ensure scalability and resilience in large-scale IIoT deployments? RQ3: What are the trade-offs of deploying the system in resource-constrained IIoT environments? By addressing these questions, this research seeks to bridge critical gaps in the literature, advancing scalable and robust trust management solutions for secure and adaptive IIoT systems.

3. Dynamic trust management system

This section presents the AI-DTMS architecture, which uses machine learning to evaluate IoT device data trustworthiness. As shown in Fig. 1, AI-DTMS assesses trust through metrics analyzing device interactions and communication patterns. Raw data, combined with static and dynamic attributes, informs reputation analysis by comparing observed and historical trust. An SVM model classifies devices as trusted or untrusted, with trust scores updated dynamically based on real-time behavior and new device integration. Detailed discussion is given below:

3.1. Collection of knowledge

The process of gathering relevant information, data, and insights necessary for effective trust evaluation. The collected knowledge forms the foundation upon which trust decisions are made, influencing the system's ability to assess and manage trust effectively.

Knowledge collection starts by identifying diverse information sources that contribute to trust evaluation. These sources include data types such as the state of IoT devices (sensors and actuators), which offer critical insights into device behavior, performance, and reliability for trust assessment. Network statistics, including bandwidth utilization, throughput, and latency, are another key source, providing measurements related to network performance and communication dynamics. Furthermore, factors such as environmental parameters (e.g., temperature, humidity), and situational contexts (e.g., time of day, user interactions) provide valuable context for understanding device behavior and performance.

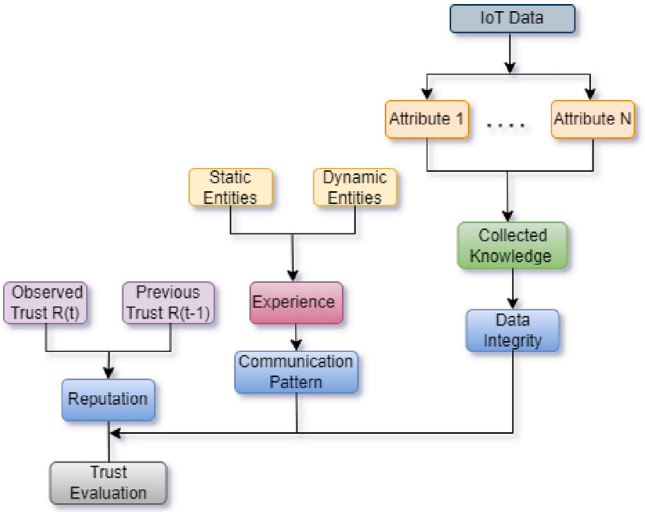


Fig. 1. Trust evaluation metrics of the system.

Data Preprocessing: Upon acquisition, collected attributes undergo preprocessing to prepare them for trust evaluation. As part of this process, dynamic weights are applied to the attributes to dynamically adjust their importance over time. The dynamic weight adjustment is governed by the formula (Eq. (1)):

$$w_i(t) = \frac{1}{1 + e^{-\alpha_i(t-t_0)}} \tag{1}$$

Where $w_i(t)$ represents the dynamic weight of attribute i at time t, α_i controls the rate of adjustment, and t_0 is the reference time. This dynamic weighting mechanism adapts to the evolving nature of IIoT devices by adjusting the importance of attributes based on changing device behaviors, communication patterns, and historical data. This ensures that the trust evaluation process remains responsive to shifts in network conditions over time. The system dynamically prioritizes attributes based on their current relevance and significance, with features indicative of trustworthiness receiving higher weights. This flexibility allows the TMS to continuously evolve alongside the devices and their interactions within the network.

Additionally, the sigmoidal function in Eq. (1) facilitates smooth, continuous adjustments to attribute weights over time. The sigmoidal curve enables gradual weight changes, preventing abrupt fluctuations that could destabilize the trust evaluation. This ensures a stable and consistent approach to attribute weighting, even when handling noisy or fluctuating data. Our system uses five key attributes: Device Data (D_{d_i}), Payload Size (P_{d_i}), Latency (L_{d_i}), Bandwidth (B_{d_i}) and Timestamp (TS_{d_i}), as a knowledge base for trust decisions. The dynamic weight adjustment mechanism selectively adjusts the weights of critical attributes, including P_{d_i}, L_{d_i}, and B_{d_i}, based on the communication interval (Δt_i). These attributes are crucial for assessing device trustworthiness, ensuring reliable data transmission across varying communication intervals.

3.2. Learning from experience

For an effective TMS, learning from past device behavior is essential. Our system adopts a collective approach, combining static historical data with dynamic attributes to derive key metrics: Reputation R_{d_i}, Integrity I_{d_i}, and Communication Pattern Analysis C_{d_i}. By integrating historical and real-time data, the system ensures a holistic assessment of device trustworthiness.

Reputation Calculation: Reputation R_{d_i} is a critical parameter that quantifies the overall trust value associated with the device, derived from its behavior history. It is calculated as:

$$R_{d_i} = 1 - |1 - \mu_{d_i\tau}| \tag{2}$$

where $\mu_{d_i\tau}$ represents the mean overall trust value associated with a device d_i. It represents a device-specific measure of trust based on its behavior history. R_{d_i} evaluates the deviation of a device's trust value from the mean overall trust value, indicating its relative reputation within the network. A R_{d_i} score nearing 1 indicates a device's consistent and reliable behavior, signifying high trustworthiness. In contrast, a score closer to 0 highlights inconsistent or unreliable behavior, raising concerns about the device's trust.

Integrity Calculation: I_{d_i} provides a comprehensive assessment of a device's communication integrity based on key attributes such as P_{d_i}, L_{d_i}, and B_{d_i}. By incorporating Integrity as a key parameter in the trust evaluation process, our system promotes reliable and consistent communication behavior among networked devices. Our system calculates I_{d_i} as:

$$I_{d_i} = \alpha \times P_{d_i} + \beta \times L_{d_i} + \gamma \times B_{d_i} \tag{3}$$

Where I_{d_i} aggregates the contributions of P_{d_i}, L_{d_i}, and B_{d_i}. The weight coefficients α, β, and γ represent the relative importance of each driven attribute in determining a device's integrity.

Communication Pattern Analysis: C_{d_i} is a critical parameter in evaluating the stability and predictability of a device's communication patterns within the network. It assesses the stability of Δt_i between consecutive communications, reflecting the device's ability to maintain consistent communication behavior over time. The Communication Stability is calculated as:

$$C_{d_i} = f_{\text{data}} \times S_{\text{comm}} \tag{4}$$

Where f_{data} represents the frequency of data transmission, reflecting how often a device communicates within the network, and S_{comm} denotes the communication stability, defined as the inverse of the mean communication interval $\mu_{\Delta t}$ between consecutive communications. The f_{data}, S_{comm}, and their components are calculated as follows:

$$S_{\text{comm}} = \frac{1}{\mu_{\Delta t}} \quad \text{and} \quad \mu_{\Delta t} = \frac{1}{n}\sum_{i=1}^{n} \Delta t_i \tag{5}$$

$$f_{\text{data}} = 1 - \frac{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(\Delta t_i - \mu_{\Delta t})^2}}{\frac{1}{n}\sum_{i=1}^{n} \Delta t_i} \tag{6}$$

Here n is the total number of communication intervals taken for trust assessment. S_{comm} assesses the stability and predictability of a device's communication patterns. Its higher score corresponds to a shorter $\mu_{\Delta t}$, signifying more frequent and regular communication by the device. This indicates a higher level of stability and predictability in the device's communication behavior. Conversely, a lower score suggests longer and more irregular Δt_i, indicating potential instability or variability in the device's communication patterns.

f_{data} quantifies the device's communication activity relative to its communication intervals. By comparing the variance of Δt_i to their $\mu_{\Delta t_i}$, it offers a measure of the device's consistency and regularity in transmitting data. A higher f_{data} score indicates more stable and predictable communication behavior. Hence, by incorporating these components, C_{d_i} provides insights into the device's communication behavior over time, helping assess its reliability and consistency in transmitting data. Devices with high C_{d_i} scores demonstrate stable and reliable communication patterns, enhancing the overall stability and reliability of network systems.

3.3. Trust metrics evaluation and decision using reputation

This section evaluates trust metrics within the TMS and outlines the decision-making process. It assesses the current trustworthiness of devices based on observed metrics and determines the final trust value for decision-making.

3.3.1. Current observed trust

The current observed trust $T_{d_i}^{O}$ of each device is evaluated using a weighted combination of its R_{d_i}, I_{d_i}, and C_{d_i} parameters. The formula for calculating the current observed trust is:

$$T_{d_i}^{O} = aR_{d_i} + bI_{d_i} + cC_{d_i} \tag{7}$$

where a, b, and c are weight coefficients determining the influence of each trust parameter. $T_{d_i}^{O}$ assesses the immediate trustworthiness of devices based on their recent behavior and communication patterns. By considering R_{d_i} in the trust calculation, our system can discern devices with a proven track record of trustworthiness from those with questionable behavior. Additionally, I_{d_i} evaluates the reliability and consistency of transmitted data, which is a critical factor in ensuring the authenticity and trustworthiness of the device. Finally, the third driven value C_{d_i} represents the reliability and consistency of data transmissions, thereby enhancing the device's overall trustworthiness within the network. By assigning weights a, b, and c, these coefficients can be fine-tuned to align with specific system requirements and the relative importance of each factor. Achieving the right balance ensures an accurate and reliable representation of trust.

3.3.2. Final trust value

The final trust value $T_{d_i}^{F}$ represents the cumulative trustworthiness of a device, incorporating historical trust values and the currently observed trust. It is calculated using a weighted average of the previous trust value $T_{d_i}^{F}(t-1)$ and the current observed trust $T_{d_i}^{O}$, as shown in the formula:

$$T_{d_i}^{F} = \delta \times T_{d_i}^{F}(t-1) + (1-\delta) \times T_{d_i}^{O} \tag{8}$$

Here, δ is a parameter between 0 and 1, adjusting the influence of past trust values. By incorporating historical trust values, our system adeptly accommodates the device's long-term behavior and performance. This nuanced evaluation permits the recognition of trends and patterns within a device's trust history, offering a more insightful assessment of its trustworthiness. This weighted approach ensures a balance between past reliability and current performance. This equilibrium enables the final trust value to dynamically reflect a device's historical reliability and recent performance, offering a robust metric for trust decisions within the network.

3.4. AI-based trust prediction model and retraining

Our system employs an AI-based approach to predict the trustworthiness of device data by integrating real-time data features and historical trust metrics for informed decision-making. The system extracts relevant features from the received data and uses Support Vector Machine (SVM) algorithms to classify it as either trusted or untrusted. SVM excels in binary classification by finding an optimal hyperplane to separate data, and by training on labeled historical datasets, the system learns to identify patterns for accurate trust predictions. The robustness, flexibility, and theoretical underpinnings of SVM make it a compelling choice for trust prediction tasks.

Feature Selection, Extraction, and Prediction: In the feature selection process, we selected key features, including contextual integrity, privacy sensitivity, bandwidth usage, latency, communication interval, reputation, integrity, and communication stability, based on domain knowledge and empirical observations. Additionally, our model captures device data, including sensor or actuator status, P_{d_i}, and timestamp information TS_{d_i}. Subsequently, derived features such as L_{d_i}, B_{d_i}, and Δt_{d_i} are calculated based on the captured data. Following this, additional derived features R_{d_i}, I_{d_i}, and $C_{S_{d_i}}$ are computed using techniques discussed in the previous section to enhance the trust prediction model's accuracy and robustness. Together, these derived features offer a comprehensive understanding of device behavior and facilitate informed trust predictions. Finally, the trust score $T_{d_i}^F$ for the device is calculated by aggregating these multiple trust-related metrics, resulting in a set of input features prepared for prediction.

Decision-Making Process: To formalize the decision-making process, we introduce two fixed threshold values: τ_H and τ_L. These thresholds are real numbers and are determined through empirical analysis or predefined criteria. The low trust threshold sets a minimum trust score below which a device is considered untrusted when the machine predicts it to be untrusted. Conversely, the high trust threshold sets a value above which a device is considered trusted when the machine predicts it to be trusted. Given that F_{d_i} is the final decision for a device d_i:

$$F_{d_i} = \begin{cases} \text{trusted} & \text{if } (S_{d_i} = 1 \text{ and } T_{d_i}^F > \tau_H) \text{ or } (S_{d_i} = 0 \text{ and } T_{d_i}^F \geq \tau_L) \\ \text{untrusted} & \text{if } (S_{d_i} = 1 \text{ and } T_{d_i}^F \leq \tau_H) \text{ or } (S_{d_i} = 0 \text{ and } T_{d_i}^F < \tau_L) \end{cases} \tag{9}$$

Where $S_{d_i} \in \{0, 1\}$ is the machine prediction for the device d_i, and $\tau_H \in \mathbb{R}$ and $\tau_L \in \mathbb{R}$ are the high and low trust thresholds, respectively. This decision logic strengthens evaluation by incorporating both the machine learning model's prediction and the trust score. The thresholds play a key role in enhancing reliability and minimizing false positives and negatives.

Model Retraining: To adapt to evolving IoT devices and emerging cybersecurity threats, our system ensures continuous resilience by periodically updating the dataset and retraining the SVM model. This allows the model to integrate new attack patterns and device behaviors, maintaining its ability to detect threats in real time. However, the frequency of retraining can pose challenges, particularly in scenarios where IIoT data is updated frequently. To address this issue and optimize the retraining process, several strategies can be considered such as Continuous Learning, Update After Batch of Readings, Update After a Certain Number of Readings, and Update After Significant Reading. Each strategy was evaluated based on computational efficiency, model accuracy, and responsiveness to device behavior changes to align with application requirements.

4. Blockchain-based security system

The Blockchain System is a central component connecting the IIoT system and AI-DTMS, utilizing blockchain to store tamper-proof records of data transactions for transparency, immutability, and security. The system supports multiple organizations, geographically separated, sharing IIoT data through a blockchain network. Key functionalities include:

4.1. Registration of node and its devices

The process begins with the registration of IoT nodes and their associated devices. The network's Admin, enrolled during setup, oversees this registration. To prevent malicious device replacement, the registration process commences with the registration of an IIoT node, following Algorithm 1. The system uses the unique Media Access Control (MAC) address as an inherent identifier to strengthen its defense mechanisms. A smart contract is deployed to generate a secure signature using the tamper-resistant ECDSA algorithm, renowned for its resistance to tampering. An authenticated user provides the node name and MAC address, triggering the generation of a signature, which is then returned for future verification.

Algorithm 1 IIoT-Node Registration

Input Parameters: node_i = Node ID, mac_{node_i} = Node MAC Address
Output Parameters: sig_{node_i} = Node Signature
begin:
  If (isExist(node_i, blockchain) = false) **then**
    key_pair = generateKeyPair() // Generate public--private key pair
    createSig(node_i, mac_{node_i}, key_pair.private_key) // Create the signature
    blockchain_iot_node, key_pair.public_key // Store credentials to blockchain
    return sig_{node_i}
  else return // This node node_i is already registered
  end

Registration of Devices: Following the registration of IoT nodes, each associated device (e.g., sensors) is registered using the node's ID and signature, as outlined in Algorithm 2. The process begins by verifying the node's authenticity. Upon successful authentication, the system checks whether the device is already registered. If it is not, a unique device ID in the format 'Node_id.Device_id' is generated and registered.

Algorithm 2 Node-Device Registration

Input: node_i (Node ID), device_i (Device ID), sig_{node_i} (Node Signature)
Output: reg_{device} (Registered Device ID)
begin:
  if `isExist`(node_i, blockchain) **and** sig_{node_i} = verified **then**
    if `isExist`(device_i, blockchain) = false **then**
      $\text{reg}_{device} \leftarrow \text{node}_i.\text{device}_i$
      `register_device`(reg_{device}, blockchain)
      return reg_{device}
    else return "`Device already registered`"
  else return "`Node authentication failed`"
end

4.2. Signature generation and verification technique

The model employs ECDSA for signing the IoT node's MAC address. ECDSA with the P256 curve is chosen due to its 128-bit security and efficient performance, making it suitable for IoT applications. At registration, the user inputs the IoT node's MAC address (M) and node ID (N). The MAC address is hashed using SHA256 to create a secure hash value (H) using the `SHA256` algorithm. This algorithm is widely used for irreversible and unique hash generation with a 256-bit digest size security.

$$H = \text{SHA256}(M)$$

Deterministic Random Number Generation Method: During the generation of ECDSA key pairs, the challenge arose of creating a large random number that could vary across peers. Since the Hyperledger Fabric framework requires identical signatures among endorsing peers, this randomness posed a significant issue. To overcome this, a deterministic approach was adopted, using the MAC address hash as a seed for the security parameter. This ensures consistency across peers, reducing endorsement errors and computational overhead. The modified process for security parameter generation is as follows: k = Hash(MAC_Address).

Thereafter, private key (d) and public key (Q) pairs are generated, and the signature is created using d, H, and the modified random number k. After collecting the required parameters, we sign M using d, H, and k: Signature (r, s): (k, d, H) where:

$$r = x \mod n, \quad s = k^{-1}(H + dr) \mod n$$

Here, (x, y) is a point on the elliptic curve, and n is the curve's order. The values Q, N, and H are stored in the state database for later verification: Database entry: (Q, N, H). Before returning the signature to the user, these are encoded in string format:

Returned signature: $S = \text{encode}(r, s)$

Verification of signature

Once IoT nodes are registered, transactions can be performed by verifying node signatures. The user enters N and S. The corresponding stored hash H and Q are retrieved from the database. Before verification, the encoded signature is first decoded to its initial format, such as from string to byte: *Signature: (r, s) = decode(S)*. Verification follows the standard ECDSA process, ensuring authenticity and integrity. As the process remains unchanged, further details are omitted for brevity. If verified, the sensor is associated with the node, and the transaction proceeds; otherwise, it is rejected.

5. System architecture and implementation

This section provides an overview of the architecture and its constituent elements, highlighting their functionalities and interactions within the broader ecosystem (See Fig. 2).

5.1. System architecture

Our system architecture comprises three main modules: the IIoT system, TMS, and the Blockchain System. Additionally, we have the Admin for administrator work and the Client or Org Operator interface, which serves as the interface for interacting with the system.

IIoT System with Organizations: The IIoT devices – sensors and actuators – are distributed across organizational units (e.g., Org1, Org2). Sensors collect real-time data, which is transmitted to the blockchain for processing and storage, while actuators execute commands to control industrial processes.

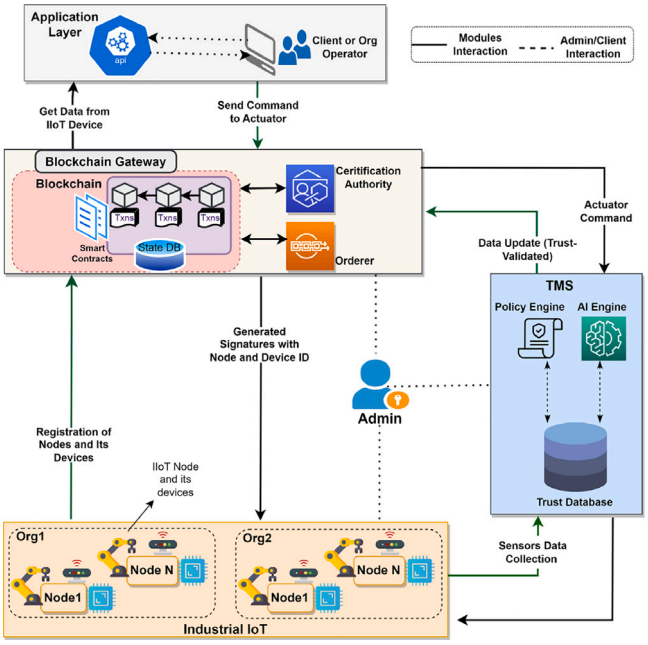


Fig. 2. IIoT device data transaction flow 1.

Application Layer: The admin, with elevated privileges, oversees the system's operation, ensuring smooth functionality. The Client or Org Operator interface offers a user-friendly platform for data retrieval, actuator control, and communication with external entities.

Trust Management System: The TMS ensures data integrity and security. It includes a *Policy Engine* that enforces policies based on trust scores, an *AI Engine* that dynamically updates trust scores using machine learning, and a *Trust Database* that stores these scores to support informed decision-making.

Blockchain System: The blockchain system registers nodes and devices, ensuring secure and authenticated setup. It uses signatures to validate data transactions, storing them in a decentralized, immutable ledger to preserve integrity. Key components include transaction-validating nodes, smart contracts for automation and compliance, a state database for transaction records, a certification authority for device authenticity, and an orderer to maintain transaction consistency.

5.2. Components interaction

This section outlines the coordinated operation of system components and their interactions within the architecture to ensure smooth data exchange, trust management, and system administration. Key steps are given as follows:

5.2.1. Registration of IIoT nodes and devices on TMS

Before IIoT devices (sensors and actuators) can interact within the blockchain system, they must first be registered. This process uses an ECDSA-based digital signature for device authentication (as detailed in Section 4). Once registered, devices can transmit data via the AI-DTMS, but must first be registered within the TMS also. Upon the device's first data transmission, the TMS checks its database for the device's presence. If not found, it requests verification from the blockchain system. The blockchain validates the device's signature, and the TMS updates its database, registering the device for future verification. For subsequent communications, the TMS authenticates the device's identity by matching its ID and signature with the database records. If a match is found, the system proceeds to transmit the data to the blockchain, updating the state database.

5.2.2. Communicating IIoT devices data

Data communication in the system is secured through HTTPS, leveraging TLS encryption to protect against eavesdropping, tampering, and man-in-the-middle attacks. This ensures the integrity and confidentiality of interactions between the IIoT system, TMS, and blockchain platform. When an IoT device transmits data, the TMS first validates the details against the Trust Database (TDB). If it is the device's first communication, the TMS forwards the device's details to the blockchain system for verification. After successful validation, the blockchain sends confirmation to the TMS, which then stores the device's ID and signature in the TDB.

For subsequent interactions, the TMS authenticates the device by matching its ID and signature in the TDB before performing a trust assessment. This assessment follows Algorithm 3, analyzing sensor data consistency, anomaly detection, and historical patterns. Key information, including Node Signature, Device ID, and Timestamp, is input to check if the node is in the database. If it is the first

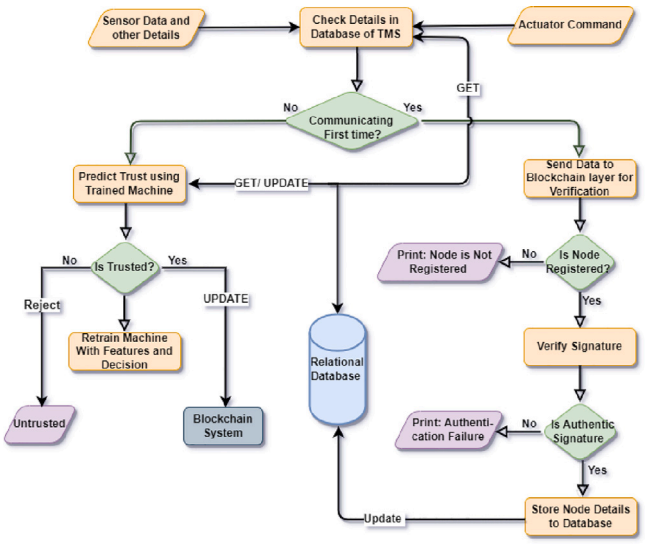


Fig. 3. IIoT device data transaction flow.

communication, the device is registered as outlined earlier. For subsequent communications, trust factors such as P_{d_i}, Δt, L_{d_i}, and B_{d_i} are calculated, and weights are dynamically adjusted based on evolving conditions. Utilizing the basic and computed features, the system derives features such as R_{d_i}, I_{d_i}, and C_{d_i}. These factors contribute to the overall trust score for the device, which the trained machine learning model uses to predict trustworthiness. If the data is trustworthy, it is integrated into the blockchain. Otherwise, the system rejects it. This process is visualized in Fig. 3.

Algorithm 3 Trust Calculation Process

Input Parameters: device_i = Device ID, sig_{device_i} = Node Signature, TS_{d_i} = Timestamp
Output Parameters: F_{d_i} = Trusted/Untrusted
begin:
$\text{P}_{d_i}, \text{L}_{d_i}, \text{B}_{d_i} \leftarrow$ `calculate_basic_features`$(\text{D}_{d_i}, \text{TS}_{d_i})$
$\text{w}_{d_i}(\text{t}) \leftarrow$ `calculate_dynamic_weight`$(\text{D}_{d_i}, \text{P}_{d_i}, \text{L}_{d_i}, \text{B}_{d_i}, \Delta\text{t}_{d_i})$
$\text{R}_{d_i}, \text{I}_{d_i}, \text{C}_{d_i} \leftarrow$ `calculate_derived_features`$(\mu^{r}_{d_i}, \text{P}_{d_i}, \text{L}_{d_i}, \text{B}_{d_i}, \Delta\text{t}_{d_i})$
$\text{T}^{O}_{d_i} \leftarrow$ `calculate_current_trust`$(\text{R}_{d_i}, \text{I}_{d_i}, \text{C}_{d_i})$
$\text{T}^{F}_{d_i} \leftarrow$ `calculate_overall_trust`$(\text{T}^{F}_{d_i}(\text{t-1}), \text{T}^{O}_{d_i})$
$\text{S}_{d_i} \leftarrow$ `predict`$(\text{D}_{d_i}, \text{P}_{d_i}, \text{L}_{d_i}, \text{B}_{d_i}, \text{TS}_{d_i})$
if $\text{F}_{d_i}(\text{S}_{d_i}, \text{T}^{F}_{d_i}) = \text{trusted}$ **then**
  blockchain $\leftarrow \text{device}_i, \text{D}_{d_i}$
end

5.3. System setup and configuration

The IIoT system forms the foundation of the framework, responsible for collecting sensor data from manufacturing units. IoT nodes, powered by Raspberry Pi 3 devices, handle real-time data processing, while DHT and Ultrasonic sensors capture environmental data like temperature, humidity, and distance. Node-RED is used to integrate and manage sensor data within the IoT ecosystem (see Table 1). The AI-DTMS ensures data integrity and evaluates the trustworthiness of incoming data. It incorporates a Python-based Artificial Learning Model using SVM with an RBF kernel. The SVM model was initially trained with $\gamma = 0.2$ and C=1, parameters optimized for balance between complexity and performance. Additional initial values are detailed in Table 2. Trust data is stored in a MySQL database, serving as the TDB, and secure HTTPS communication protocols ensure confidential data exchange. To train the SVM model, we generated a dataset with the Contiki-NG simulation tool, comprising over 9000 transaction records from 1000 virtual IoT nodes. The dataset includes 25 malicious nodes simulating attacks such as DoS and Reply, enabling the model to learn from both normal and malicious behaviors, enhancing prediction accuracy.

Blockchain, utilizing HF 2.4, maintains a tamper-proof record of data transactions with modular architecture and multi-channel support for isolated communication between devices or organizations. Chaincode, written in Go, enforces business logic within the blockchain network, ensuring transparency. Docker containers deploy and manage blockchain services for scalability and consistency across environments. The experimental setup includes three virtual machines representing distinct organizations, running on a system with an Intel i7 processor, 12 GB RAM, and 512 GB SSD, connected to a high-speed 5G network. Table 2 provides details on the tools used in the experiment.

Table 1
Initial values and assumptions.

Parameter	Value
Trust Weights	$R_d^i = 1, I_d^i = 1, C_d^i = 0.5$
Feature Weights	$D_d^i = 1.5, P_{d_i} = 1, L_{d_i} = 1.5, B_{d_i} = 1, \Delta t = 0.5$
Dynamic Weight	$Q_i = 1.5$
Trust Threshold	$\tau_H = 0.5, \tau_L = -0.5$

Table 2
Main software used.

Software	Version
HF, Go Lang.	2.4.0, go1.20
Docker Engine	20.10.7
Docker Composer	1.17.1
Virtual Machines	Ubuntu 22
NPM	6.14.15

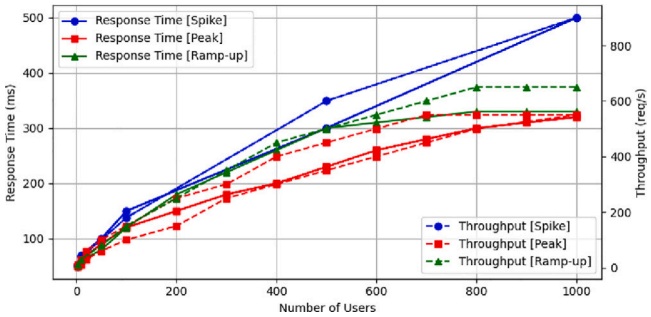


Fig. 4. Throughput and response time Vs loads.

6. Result and analysis

In this section, we explain the outcomes obtained from the implementation test scenarios which are based on performance, security and ability to classify data as trusted or untrusted.

6.1. Performance evaluation

The system was tested under varying load conditions (spike, peak, ramp-up) using throughput and response time metrics (Fig. 4). Response time increased from 210 ms (4 users) to 1350 ms (1000 users), with 90th/95th percentiles peaking at 1400 ms and 1450 ms. Throughput scaled linearly, peaking at 450 req/s for 1000 users but saturating beyond 500 users (Fig. 5). Bandwidth analysis showed response time improving from 200 ms (1 Mbps) to 90 ms (200 Mbps), while higher latency (20 ms to 150 ms) increased response time from 100 ms to 220 ms and reduced throughput from 120 req/s to 60 req/s (Fig. 6). Fig. 7 shows CPU usage peaked at 9%, while memory reached 30%, demonstrating efficient resource utilization across 4–1000 users. Scalability tests confirmed support for 1000+ users, stable performance under stress, and consistent adaptation to bandwidth and latency variations. The blockchain module was evaluated using well known Hyperledger Caliper tool. Key functions tested included *regOrgNode*, *regOrgDevice*, *setOrgData*, and *getOrgCommand*. With batch timeout, block size, and transaction duration constant, tests at 120 TPS per client (2 clients, 1 sensor each) recorded maximum, minimum, and average latencies of 1.03 s, 0.08 s, and 0.63 s, respectively. Subsequent rounds tested transaction rates from 50 to 250 TPS. Throughput and latency showed initial improvements with increasing transactions but plateaued at higher rates due to resource constraints. At higher loads, throughput and latency exhibited near-linear behavior (see Figs. 8, 9).

These results highlight the system's efficient resource management, its ability to maintain stable performance under varying loads and network conditions, and confirm the module's scalability and reliability in handling diverse transaction loads in dynamic IoT environments.

6.2. Trust evaluation performance and bias mitigation

The AI-DTMS for IIoT devices was first evaluated using a dataset of 9641 instances, covering various devices and network scenarios. Performance was assessed using accuracy, precision, recall, and F1 score. The confusion matrix (Fig. 12) showed 5190 true

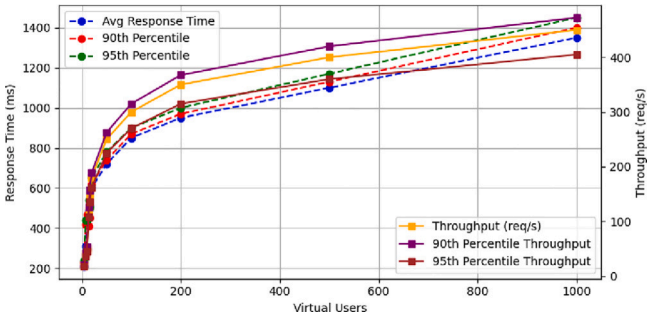


Fig. 5. Throughput and response time percentile.

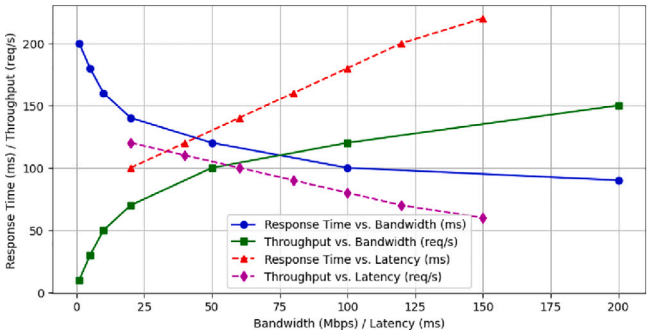


Fig. 6. Performance vs bandwidth/latency.

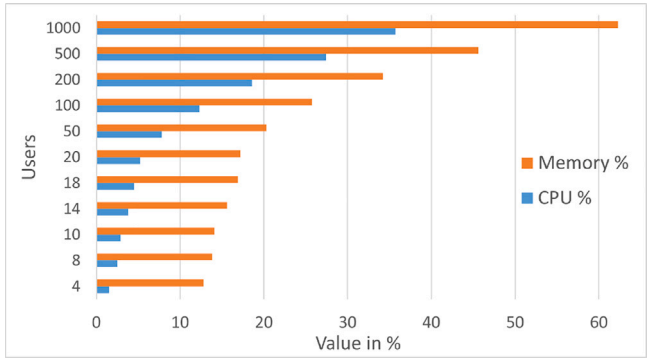


Fig. 7. CPU and memory usage.

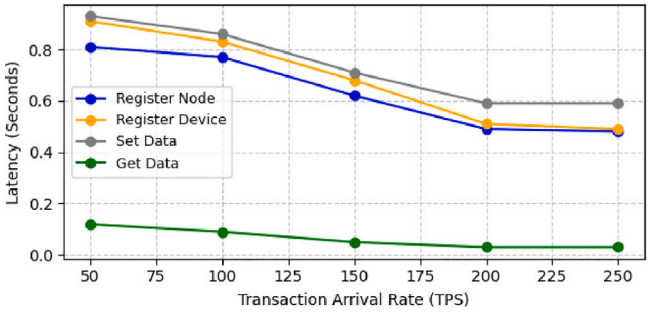


Fig. 8. Latency of functions.

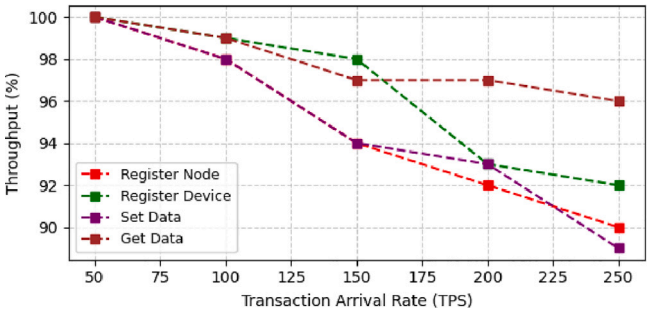


Fig. 9. Throughput of the functions.

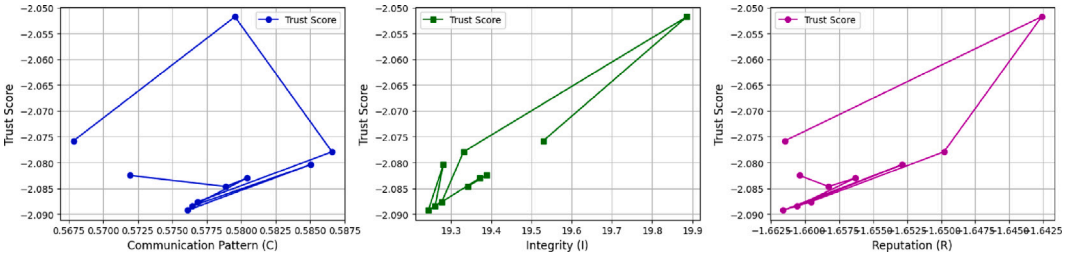


Fig. 10. Communication pattern, integrity and reputation vs trust score.

positives (TP), 4451 true negatives (TN), 385 false positives (FP), and 23 false negatives (FN), demonstrating strong classification performance. The system achieved an accuracy of 96.31%, precision of 93.09%, recall of 99.59%, and an F1 score of 96.23%, indicating its effectiveness in trust assessment. The low false positive and false negative rates confirm the system's ability to evaluate trustworthiness accurately in IIoT environments, ensuring security and system integrity.

To mitigate biases in trust evaluation, several techniques were employed to ensure fairness and robustness. Scaling and normalization standardized the input features, preventing any single feature from dominating the trust classification. Careful feature selection focused on unbiased factors like device behavior, communication frequency, and data integrity. The diverse and balanced dataset included various IIoT devices, manufacturers, and network scenarios, minimizing overfitting and bias toward specific profiles. The SVM model is periodically re-trained with new data to adapt to evolving IIoT conditions, while its kernel flexibility helps model complex, nonlinear relationships, further reducing bias. These strategies ensure reliable trust assessment and bias mitigation.

The evaluation of the AI-DTMS also involved analyzing trust metrics (R, I, C, and overall Trust Score) for various devices, along with the impact of varying adjustment rates (Q_i) on the dynamic weight function and resulting trust scores. Fig. 10 shows that while the I and C scores remain stable, R scores fluctuate slightly, affecting the overall Trust Score. These scores assess device reliability based on behavior and interactions. The dynamic weight function adjusts feature weights over time, with the impact of different Q_i values on $W(t)$ and $T_{d_i}^F$ evaluated under a 10 ms transaction interval (Fig. 11). The graph demonstrates that lower Q_i values result in smaller adjustments in $W(t)$ and less fluctuation in $T_{d_i}^F$, while higher Q_i values cause more significant changes.

6.3. Security and resilience assessment

This section discusses possible security attacks on operational industrial IoT applications and the resiliency of our model against them.

6.3.1. Authentication and MAC address integrity evaluation

The authentication success rate (ASR) evaluation focused on assessing the system's ability to successfully authenticate IoT nodes based on different signature factors. A dataset consisting of valid signatures, invalid signatures, tampered signatures, and expired signatures was used to test the authentication process. The ASR is the percentage of successful authentications, calculated as the percentage of successful authentications relative to total attempts. For valid signatures, the system achieved a 100% ASR, successfully authenticating all legitimate IoT nodes. Invalid, tampered, and expired signatures were correctly rejected, each resulting in a 0% ASR.

In addition to the ASR evaluation, the system's security against MAC address manipulation threats was also tested. This ability of this system was tested in two scenarios. First, in the case of a modified MAC address, where an attacker attempts to alter the

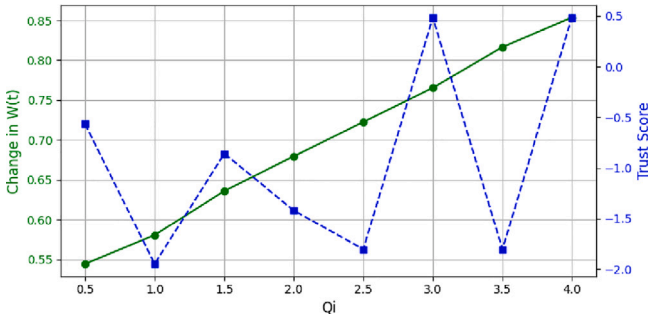


Fig. 11. Dynamic weight vs Qi.

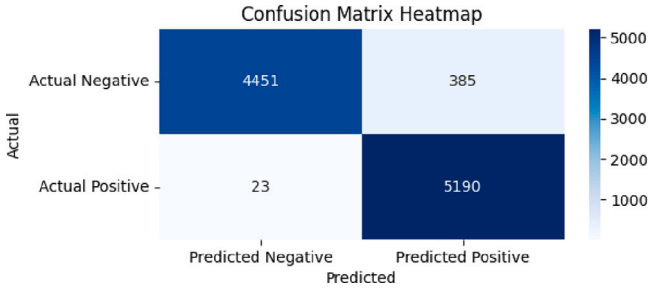


Fig. 12. Confusion matrix heat map.

Table 3
Comparison with some related existing work.

Research work	Spoofing	Sybil attacks	Node-capturing	Replay attack	DDoS attack
[9,18]	✓	×	×	×	✓
[15]	×	×	×	×	×
[12]	✓	✓	×	×	×
[13]	×	×	×	✓	✓
[10]	✓	×	×	✓	×
[11]	✓	✓	×	✓	×
Our Work	✓	✓	✓	✓	✓

MAC address of a legitimate IoT node, the system securely registers the original MAC address on the HF blockchain, detecting unauthorized changes and maintaining authenticity. Second, in the case of MAC address impersonation, where an attacker uses a false MAC address to mimic a legitimate IoT node, the system leverages the securely stored public key associated with the genuine MAC address to detect fraudulent signatures during verification. The results show that the system effectively protects against these threats while introducing minimal overhead and efficiently handles high volumes of authentication with fast response times. This confirms its scalability and suitability for real-time, large-scale IIoT applications.

6.3.2. Vulnerability and robustness testing against real-world attacks

A vulnerability assessment using OpenVAS evaluated the security of the proposed system by scanning Docker services and network infrastructure. Key findings included log-level threats on services like general/CPE-T and general/tcp for host 10.0.1.4, and medium-level threats on port general/tcp for host 10.0.1.20. Virtual Machine 1 (192.168.222.1) showed medium and low-level threats on ports 443/tcp and general/tcp, while other hosts, such as 10.0.1.18 and 10.0.1.25, exhibited log-level issues like misconfigurations. Hosts machines 192.168.222.2 and 192.168.222.3 (VMs 2 and 3) also had low or log-level vulnerabilities. No high-severity risks or false positives were detected, affirming the system's security measures and the robustness.

Furthermore, robustness testing using *Nmap* and *Metasploit* confirmed the system's resilience against targeted attacks. Multilayer authentication and chaincode-based verification effectively countered spoofing attempts, ensuring data integrity. Testing against Sybil attacks validated the robustness of registration and authentication processes, preventing fraudulent identities. Secure private key storage and signatures safeguarded against node-capturing attacks, even with physical access. TLS certificates and secure signatures mitigated replay attacks, ensuring transmitted data integrity. The system also resisted DDoS attacks through advanced authentication and transaction verification protocols, addressing device impersonation and unauthorized access. These tests validated the system's robustness and real-world applicability, as summarized in Table 3.

Table 4
Comparison between traditional approaches and proposed system.

Aspect	Traditional IIoT Security	Proposed AI-Driven TMS
Trust Evaluation	Static, rule-based policies [22]	Dynamic, real-time learning
Adaptability to Threats	Limited to predefined behaviors	Adjusts to new data and threats
ML Integration	Rare, mainly anomaly detection	Dynamic trust via SVM
Scalability	Limited by centralized bottlenecks [23]	Decentralized, modular architecture
Resilience	Vulnerable to central failures [24]	Decentralized, no single failure point
Security	Centralized, tampering-prone [25]	Blockchain-backed (e.g., HF)
Overhead	Low, increases with encryption layers	Moderate, optimized via HF consensus
Speed & Throughput	Moderate, server-dependent	High (thousands/sec) [20]
Resource Usage	Optimized for centralized tasks	Low, avoids PoW with HF
Data Storage	Centralized, tampering risk	Immutable blockchain storage
Legacy Integration	Easier with centralized setups	Achievable via middleware

6.4. Scalability analysis and performance trade-off

This section addresses RQ2 by demonstrating how the system's modular architecture ensures scalability and performance in large-scale IIoT deployments. The modular design is central to the system's scalability. The Blockchain System and AI-DTMS function independently, each with the flexibility to scale as needed. For instance, the AI-DTMS evaluates trustworthiness based on IIoT device behavior. As the number of devices increases, the TMS can scale horizontally by deploying additional compute resources or parallelizing operations, thereby managing higher volumes of trust evaluations without affecting blockchain throughput. This capability is critical in large-scale IIoT environments.

Simultaneously, the blockchain processes transactions, including trust-related data, efficiently without being impacted by the computational load from the TMS. The separation between these two components ensures fault isolation, meaning issues in one module do not affect the other. Moreover, this modularity makes it easier to upgrade the system – whether by integrating new consensus mechanisms or refining machine learning models – without disrupting operations.

To further enhance scalability, the system filters untrusted data before committing it to the blockchain. By ensuring that only trustworthy data is recorded, the system reduces unnecessary storage and processing overhead, thereby maintaining performance even as device and transaction volumes grow. Although blockchain technology offers decentralized security, it can pose performance trade-offs, particularly in large-scale environments. Traditional consensus mechanisms like PoW may introduce bottlenecks, but HF's RAFT consensus mitigates this by enabling transaction speeds of several thousand per second [20]. This efficiency is well-suited for IIoT environments that require low latency and high throughput.

By focusing the blockchain on secure, immutable data storage and assigning dynamic trust evaluation to the AI-DTMS, each module can optimize performance within its specific domain. This prevents the blockchain from becoming a limiting factor and preserves scalability. Looking ahead, while the current work addresses the scalability of the system with its existing architecture, an off-chain module developed in our subsequent work [21] will further enhance scalability by handling certain operations off-chain. This will reduce blockchain storage requirements and improve throughput, providing additional scalability benefits, particularly in large-scale deployments. By leveraging modularity, efficient consensus mechanisms, and future off-chain integration, the system effectively addresses the scalability challenges faced by IIoT networks, ensuring high performance and adaptability in dynamic, large-scale environments.

6.5. Comparative analysis

The AI-DTMS offers substantial improvements over traditional IIoT security approaches, particularly in dynamic, large-scale environments. Traditional systems rely on static, rule-based trust models that struggle to adapt to evolving threats, while AI-DTMS uses machine learning (e.g., SVMs) for dynamic, real-time trust evaluation, improving adaptability to new data and threats. Additionally, traditional systems are typically centralized, making them vulnerable to single points of failure. In contrast, the integration of blockchain, particularly with permissioned HF, decentralizes the architecture, enhancing security and resilience. HF's efficient consensus mechanism supports high throughput and low latency, ideal for IIoT environments. While the proposed system improves scalability and security, it also faces challenges. Machine learning models, such as SVMs, require high-quality data, and the blockchain may introduce latency in large-scale deployments. Additionally, integrating the system with legacy IIoT environments that rely on centralized infrastructures can be complex and costly. For a detailed comparison, refer to Table 4.

7. Conclusion

A comprehensive solution to the security and trust challenges in IIoT is achieved by integrating an AI-DTMS with a private blockchain-based security framework. The AI-DTMS dynamically assesses and predicts the trustworthiness of devices and their data, ensuring reliable network operations. The blockchain framework, utilizing modified ECDSA-based authentication and smart contracts, provides a decentralized and tamper-proof ledger, significantly enhancing security and preventing unauthorized access. The proposed system's innovative approach addresses the inadequacies of traditional security measures in dynamic IIoT environments.

Extensive simulations and evaluations confirm that our integrated approach significantly enhances trust accuracy and strengthens security. It ensures reliable performance even in large-scale deployments, effectively addressing the outlined research questions. The experimental results demonstrate that integrating AI-DTMS not only ensures robust authentication of IIoT devices and safeguards data integrity but also enhances scalability and optimizes resource utilization. By pre-processing and validating data through the TMS before committing it to the blockchain, the system reduces unnecessary overhead. Additionally, its modular design decouples trust evaluation from blockchain operations, allowing each component to scale independently and improving performance across diverse IIoT scenarios.

While the system robustly protects general data using blockchain mechanisms, it does not include specific privacy-preserving mechanisms for handling sensitive data. Future research will focus on incorporating advanced privacy-preserving techniques, such as private data collection or enhanced encryption protocols, to address this limitation. Additionally, we aim to optimize the scalability of the system, enhance the efficiency of the blockchain framework, and explore the integration of advanced AI techniques to further improve trust evaluation and security measures in diverse and large-scale IIoT networks.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] Khan A, Bourouis S, Kamruzzaman M, Hadjouni M, Shaikh Z, Laghari A, Elmannai H, Dhahbi S. Data security in healthcare industrial Internet of Things with blockchain. IEEE Sens J 2023;23:25144–51. http://dx.doi.org/10.1109/JSEN.2023.3273851.

[2] Rangappa J, Manu A, Kariyappa S, Chinnababu S, Lokesh G, Flammini F. A lightweight blockchain to secure data communication in IoT network on healthcare system. Int J Saf Secur Eng 2023;13:1015–24. http://dx.doi.org/10.18280/IJSSE.130604, Healthcare IoT Network.

[3] Sharma A, Pilli E, Mazumdar A, Jain A. BLAST-IoT: Blockchain assisted scalable trust in Internet of Things. Comput Electr Eng 2023;109:108752. http://dx.doi.org/10.1016/J.COMPELECENG.2023.108752, Scalable Trust in IoT.

[4] Xiang X, Cao J, Fan W, Xiang S, Wang G. Blockchain enabled dynamic trust management method for the internet of medical things. Decis Support Syst 2024;180:114184. http://dx.doi.org/10.1016/J.DSS.2024.114184, Dynamic Trust in Medical IoT.

[5] Feng X, Wu J, Wu Y, Li J, Yang W. Blockchain and digital twin empowered trustworthy self-healing for edge-AI enabled industrial internet of things. Inf Sci (Ny) 2023;642:119169. http://dx.doi.org/10.1016/J.INS.2023.119169, Blockchain and Digital Twin for Self-Healing IoT.

[6] Sisinni E, Saifullah A, Han S, Jennehag U, Gidlund M. Industrial internet of things: Challenges, opportunities, and directions. IEEE Trans Ind Inform. 2018;14:4724–34. http://dx.doi.org/10.1109/TII.2018.2852491, Challenges in Industrial IoT.

[7] Kumar R. A comprehensive approach to industry 4.0 security: Blockchain and dynamic access control integration. In: 2024 sixth int. conf. comput. intell. commun. technol.. 2024, p. 479–86. http://dx.doi.org/10.1109/CCICT62777.2024.00080, Industry 4.0 Security with Blockchain.

[8] Ali S, Li Q, Yousafzai A. Blockchain and federated learning-based intrusion detection approaches for edge-enabled industrial IoT networks: a survey. Ad Hoc Netw. 2024;152:103320. http://dx.doi.org/10.1016/J.ADHOC.2023.103320, Intrusion Detection in IoT Networks.

[9] Mrabet H, Alhomoud A, Jemai A, Trentesaux D. A secured industrial Internet-of-Things architecture based on blockchain technology and machine learning for sensor access control systems in smart manufacturing. Appl Sci 2022;12:4641. http://dx.doi.org/10.3390/APP12094641, Blockchain and ML for IoT Access Control.

[10] Nita S, Mihailescu M. Elliptic curve-based query authentication protocol for IoT devices aided by blockchain. Sensors 2023;23:1371. http://dx.doi.org/10.3390/S23031371, Elliptic Curve Authentication for IoT.

[11] Khashan O, Khafajah N. Efficient hybrid centralized and blockchain-based authentication architecture for heterogeneous IoT systems. J King Saud Univ - Comput Inf Sci 2023;35:726–39. http://dx.doi.org/10.1016/J.JKSUCI.2023.01.011, Hybrid Authentication Architecture for IoT Systems.

[12] Khalid U, Asim M, Baker T, Hung P, Tariq M, Rafferty L. A decentralized lightweight blockchain-based authentication mechanism for IoT systems. Clust Comput 2020;6. http://dx.doi.org/10.1007/s10586-020-03058-6, Decentralized Blockchain Authentication for IoT.

[13] Almadhoun R, Kadadha M, Alhemeiri M, Alshehhi M, Salah K. A user authentication scheme of IoT devices using blockchain-enabled fog nodes. In: Proc. IEEE/ACS int. conf. comput. syst. appl. AICCSA. 2019, p. 1–8. http://dx.doi.org/10.1109/AICCSA.2018.8612856, Blockchain-Enabled Fog Nodes for IoT Authentication.

[14] Zhang C, Li W, Luo Y, Hu Y. AIT: An AI-enabled trust management system for vehicular networks using blockchain technology. IEEE Internet Things J 2021;8:3157–69. http://dx.doi.org/10.1109/JIOT.2020.3044296, AI-Enabled Trust Management in Vehicular Networks.

[15] Kouicem D, Imine Y, Bouabdallah A, Lakhlef H. A decentralized blockchain-based trust management protocol for the internet of things. IEEE Trans Dependable Secur Comput 2020;5971:1–14. http://dx.doi.org/10.1109/TDSC.2020.3003232, Blockchain-Based Trust Management for IoT.

[16] Wu D, Ansari N. A trust-evaluation-enhanced blockchain-secured industrial IoT system. IEEE Internet Things J 2021;8:5510–7. http://dx.doi.org/10.1109/JIOT.2020.3030689, Blockchain-Secured Trust Evaluation for Industrial IoT.

[17] Zhang F, Wang H, Zhou L, Xu D, Liu L. A blockchain-based security and trust mechanism for AI-enabled IIoT systems. Futur Gener Comput Syst 2023;146:78–85. http://dx.doi.org/10.1016/J.FUTURE.2023.03.011, Blockchain-Based Trust for AI in IIoT.

[18] Xu X, Wei S, Zhang Z, Lv P. Main-secondary blockchain framework: Cross-domain trust management mechanism using trust ticket. IEEE Trans Netw Serv Manag 2023;20:2830–44. http://dx.doi.org/10.1109/TNSM.2023.3236673, Cross-Domain Trust Management Using Blockchain.

[19] Wu X, Liu Y, Tian J, Li Y. Privacy-preserving trust management method based on blockchain for cross-domain industrial IoT. Knowledge- Based Syst 2024;283:111166. http://dx.doi.org/10.1016/J.KNOSYS.2023.111166, Privacy-Preserving Trust Management for IoT.

[20] Guggenberger T, Sedlmeir J, Fridgen G, Luckow A. An in-depth investigation of the performance characteristics of hyperledger fabric. Comput Ind Eng 2022;173:108716. http://dx.doi.org/10.1016/j.cie.2022.108716.

[21] Kumar R, Sharma R. Enhancing scalability and performance in IoT networks through off-chain data processing and blockchain integration. In: 2024 IEEE int. conf. commun. comput. signal process.. 2024, p. 1–6. http://dx.doi.org/10.1109/IICCCS61609.2024.10763678.

[22] Wang J, Jing X, Yan Z, Fu Y, Pedrycz W, Yang L. A survey on trust evaluation based on machine learning. ACM Comput Surv 2020;53. http://dx.doi.org/10.1145/3408292.
[23] Jamil M, Schelen O, Monrat A, Andersson K. Enabling industrial internet of things by leveraging distributed edge-to-cloud computing: Challenges and opportunities. IEEE Access 2024;12. http://dx.doi.org/10.1109/ACCESS.2024.3454812.
[24] Mirani A, Velasco-Hernandez G, Awasthi A, Walsh J. Key challenges and emerging technologies in industrial IoT architectures: A review. Sensors 2022;22:1–31. http://dx.doi.org/10.3390/s22155836.
[25] Fun T, Samsudin A. Recent technologies, security countermeasures and ongoing challenges of industrial internet of things (iIoT): A survey. Sensors 2021;21:1–30. http://dx.doi.org/10.3390/s21196647.

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